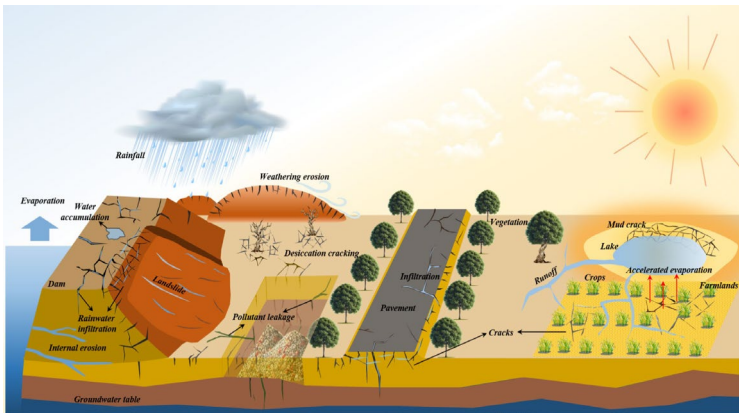


非饱和黏土弯液面演化的数值模拟与表面张力作用机制

研究背景

- 全球变暖加剧，黏性土体的收缩和开裂已经成为诱发各种次生灾害的重要因素，给众多工程领域带来了巨大的困扰。



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- 开裂现象在黏性土或膨胀土中表现得更为显著。
- 相对于饱和土体，非饱和土的失水收缩机制更为复杂。

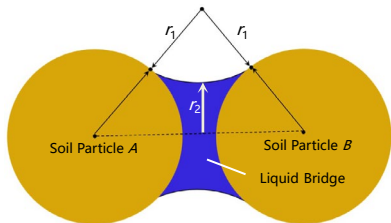


(a) 2022年美国得克萨斯州的黏质土体开裂



(b) 2024年10月非洲摩洛哥王国的沙质土体开裂

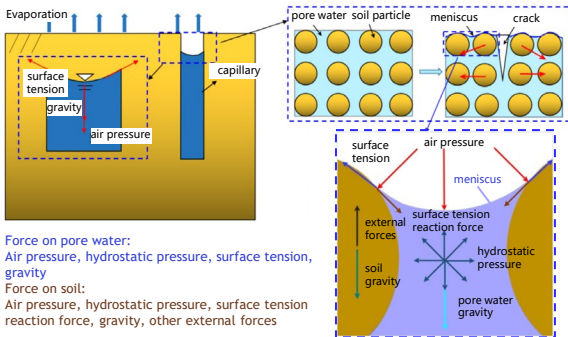
黏土干缩开裂的微观解释



The matrix suction of unsaturated soil is defined by Young-Laplace Equation, related to the morphology curved liquid surface

$$s = \gamma_s \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

Micro-scale meniscus evolution controls capillary suction → governs macro-scale shrinkage and cracking!

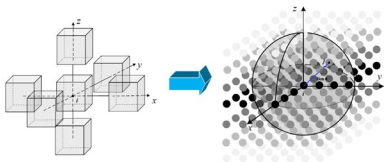


研究现状

有关弯液面的数值建模方法

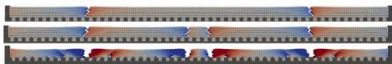
研究方法	模型	特点	局限性
DDA	一般模型	土体被离散为不规则的多边形颗粒	弯液面的固液接触角和曲率半径恒定，不能获取动态演化过程
	二维扩展模型	根据土体单位体积的含水量确定弯液面形态	
DEM	一般模型	表面张力的作用由黏附力代替	缺乏对弯液面的描述 引入了众多参数来描述弯液面的形态，可能带来较大的计算误差，且不能获取动态演化过程
	孤立毛细管模型	适用于低饱和度的二维土体	
	索道液桥模型	适用于高饱和度的二维土体	
	复杂流体团簇模型	适用于高饱和度的三维土体	
CFD-DEM	单向流固耦合模型	采用CFD方法模拟水压，采用DEM方法模拟土体颗粒运动，考虑非饱和土中的基质吸力时采用恒定粘附力	流场不受土体颗粒运动的影响，不能完整地反映表面张力作用下的流固相互作用
MD	一般模型	纳米尺度的液桥模型	基于分子力场进行建模和计算，缺乏对表面张力的描述

近场动力学在土体研究中的应用

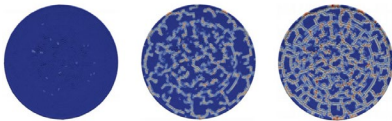


Local model of
continuum
mechanics

Non-local model
of peridynamics
(Silling, 2000)

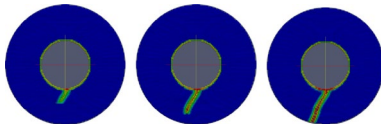


Evolution of cracking in soil strip

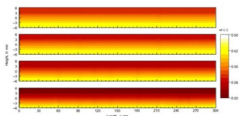


Evolution of crazing in soil plate

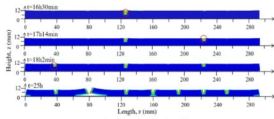
(Menon and Song, 2019)



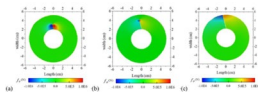
(Jabakhani, 2013)



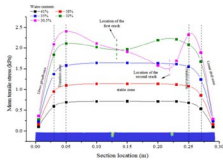
Moisture transport in soil strip



Cracking of soil strip



Evolution of stress in soil ring cracking



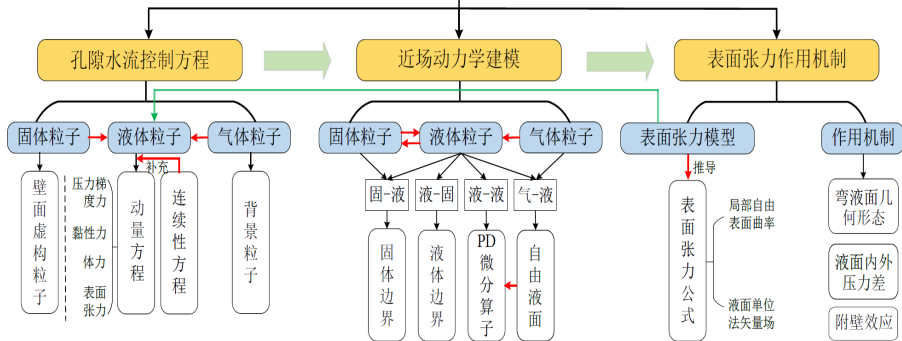
Distribution of stress in soil strip cracking

(Yan et al., 2023; Liu et al., 2023)

技术路线

非饱和黏土弯液面演化的数值模拟与表面张力作用机制研究

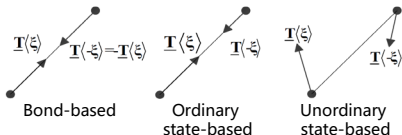
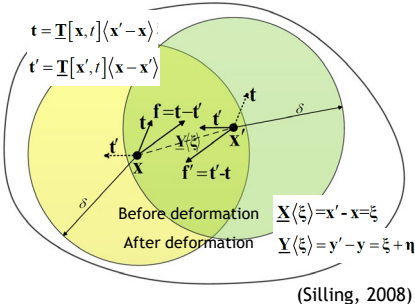
微观尺度黏土单液桥结构的简化模型



难点：计算复杂度高、规模大，边界条件施加困难！

近场动力学理论框架和微分算子

Interaction between material points
in PD modeling framework



Peridynamic differential operators

$$\frac{\partial^{n_1+n_2+\dots+n_M} f(\mathbf{x})}{\partial x_1^{p_1} \partial x_2^{p_2} \dots \partial x_M^{p_M}} = \int_{H_{\mathbf{x}}} f(\mathbf{x} + \boldsymbol{\xi}) g_N^{p_1 p_2 \dots p_N}(\boldsymbol{\xi}) dV_{\mathbf{x}'}$$

(Madenci et al., 2016)

Governing equations
of free surface flow

→
PDDO

Non-local form
expression

Lagrangian description

$$\begin{aligned} \frac{\partial \rho}{\partial t} &= -\rho \nabla \cdot \mathbf{v} \\ \rho \frac{\partial \mathbf{v}}{\partial t} &= -\nabla P + \mathbf{F}_V + \mathbf{F}_B + \mathbf{F}_S \end{aligned}$$

where

$$P = \frac{\rho_0 c_f^2}{\gamma} \left(\left(\frac{\rho}{\rho_0} \right)^\gamma - 1 \right) + P_0$$

$$\mathbf{F}_V = \mu(\Delta \mathbf{v})$$

$$\mathbf{F}_B = 0$$

$$\mathbf{F}_S = \beta \kappa \mathbf{n} \nabla s$$

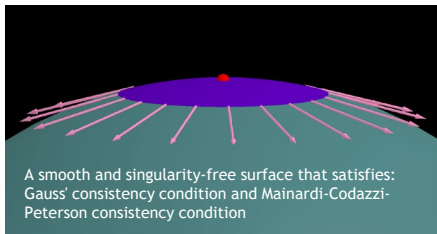
PD integral equations

$$\begin{aligned} \frac{\partial \rho(\mathbf{x})}{\partial t} &= -\rho(\mathbf{x}) \int_{H_{\mathbf{x}}} \mathbf{g}_1(\boldsymbol{\xi}) \cdot (\mathbf{v}(\mathbf{x}') - \mathbf{v}(\mathbf{x})) dV_{\mathbf{x}'} \\ \rho \frac{\partial \mathbf{v}}{\partial t} &= - \int_{H_{\mathbf{x}}} (P(\mathbf{x}') - P(\mathbf{x})) \mathbf{g}_1(\boldsymbol{\xi}) dV_{\mathbf{x}'} \\ &\quad + \int_{H_{\mathbf{x}}} \mu(\mathbf{x}) (\mathbf{v}(\mathbf{x}')) \\ &\quad - \mathbf{v}(\mathbf{x}) \text{tr}(\mathbf{g}_2(\boldsymbol{\xi})) dV_{\mathbf{x}'} \\ &\quad - \beta \int_{H_{\mathbf{x}}} \kappa(s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\boldsymbol{\xi}) dV_{\mathbf{x}'} \end{aligned}$$

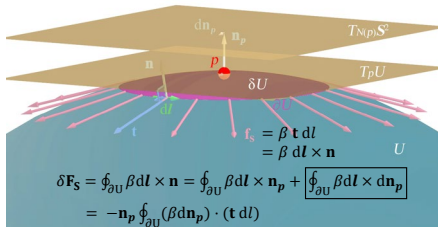
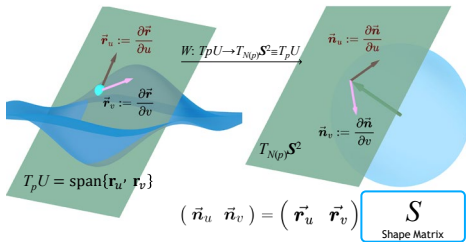
where

$$\kappa = \frac{\int_{H_{\mathbf{x}}} (\mathbf{n}(\mathbf{x}') - \mathbf{n}(\mathbf{x})) \cdot \mathbf{g}_1(\boldsymbol{\xi}) dV_{\mathbf{x}'}}{\int_{H_{\mathbf{x}}} w_0(\|\boldsymbol{\xi}\|) dV_{\mathbf{x}'} / \int_{H_{\mathbf{x}}} w_0(\|\boldsymbol{\xi}\|) dV_{\mathbf{x}'}}$$

界面流体元的表面张力模型

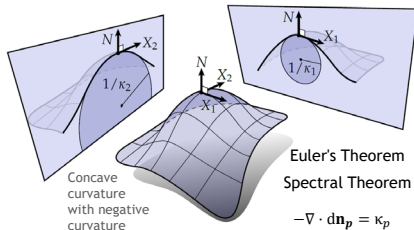


$$\vec{r}(u, v) = \begin{pmatrix} x(u, v) \\ y(u, v) \\ z(u, v) \end{pmatrix} \xrightarrow{N: U \rightarrow S^2 \subset \mathbb{R}^3} \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|} \quad \boxed{\vec{n}(u, v)} = \begin{pmatrix} n_1(u, v) \\ n_2(u, v) \\ n_3(u, v) \end{pmatrix}$$



Gauss theorem $\Rightarrow \vec{n}_p \oint_{\partial U} (\beta d\vec{n}_p) \cdot (\vec{t} dl) = \beta (\nabla \cdot d\vec{n}_p) \delta U \vec{n}_p$

$$\vec{F}_S = \lim_{\delta U \rightarrow 0} \frac{\delta \vec{F}_S}{\delta U} = \beta (-\nabla \cdot d\vec{n}_p) \vec{n}_p$$



控制方程、时间差分格式、表面张力计算

Continuity Equation

$$\rho_i^{n+1} = \rho_i^n - \rho_i^n \Delta t \sum_{j=1}^{N_i} [(\mathbf{v}_j^n - \mathbf{v}_i^n) \cdot \mathbf{g}_1(\xi_{ij}^n)] V_j$$

$$\rho_i^{n+1} = \rho_i^n - \frac{2\Delta x^2}{\pi\delta^2} \rho_i^n \Delta t \sum_{j=1}^{N_i} (\mathbf{v}_j^n - \mathbf{v}_i^n) \frac{[\xi_{ij1}^n \ \xi_{ij2}^n]^T}{\|\xi_{ij1}^n + \xi_{ij2}^n\|}$$

For two-dimensional problems, the relative value form of the near-field dynamic differential operator is adopted.

Momentum equation

$$\mathbf{a}_i^{n+1} = \frac{1}{\rho_i^{n+1}} \left(-\sum_{j=1}^{N_i} (p_j^n - p_i^n) \mathbf{g}_1(\xi_{ij}^n) V_j + \sum_{j=1}^{N_i} \mu_i (\mathbf{v}_j^n - \mathbf{v}_i^n) \text{tr}(\mathbf{g}_2(\xi_{ij}^n)) V_j \right. \\ \left. - \frac{\sum_{j=1}^{N_i} \min(N_i, N_j) (\mathbf{n}_j^n - \mathbf{n}_i^n) \cdot \mathbf{g}_1(\xi_{ij}^n) V_j}{\sum_{j=1}^{N_i} \min(N_i, N_j) w_0(\|\xi_{ij}^n\|) V_j / \sum_{j=1}^{N_i} w_0(\|\xi_{ij}^n\|) V_j} (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\xi_{ij}^n) V_j \right) + \frac{1}{\rho_i^{n+1} \mathbf{F}^B}$$

$$\mathbf{a}_i^{n+1} = \frac{2\Delta x^2}{\pi\delta^2 \rho_i^{n+1}} \left(-\sum_{j=1}^{N_i} (p_j^n - p_i^n) \frac{[\xi_{ij1}^n \ \xi_{ij2}^n]^T}{\|\xi_{ij1}^n + \xi_{ij2}^n\|} + \frac{3\mu_i}{\delta} \sum_{j=1}^{N_i} \frac{(\mathbf{v}_j^n - \mathbf{v}_i^n)}{\sqrt{\|\xi_{ij1}^n + \xi_{ij2}^n\|}} \right. \\ \left. - \frac{\sum_{j=1}^{N_i} \min(N_i, N_j) (\mathbf{n}_j^n - \mathbf{n}_i^n) \frac{[\xi_{ij1}^n \ \xi_{ij2}^n]^T}{\|\xi_{ij1}^n + \xi_{ij2}^n\|}}{\frac{\pi\delta^2}{2\beta\Delta x^2} \sum_{j=1}^{N_i} \min(N_i, N_j) w_0(\|\xi_{ij}^n\|) V_j / \sum_{j=1}^{N_i} w_0(\|\xi_{ij}^n\|) V_j} (s(\mathbf{x}') - s(\mathbf{x})) \frac{[\xi_{ij1}^n \ \xi_{ij2}^n]^T}{\|\xi_{ij1}^n + \xi_{ij2}^n\|} \right) + \frac{1}{\rho_i^{n+1} \mathbf{F}^B}$$

Time difference scheme

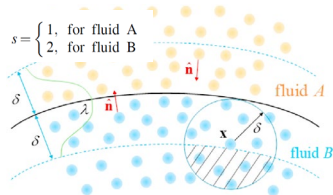
Second-order Verlet-Velocity algorithm

$$\begin{cases} \mathbf{v}_i^{n+1} = \mathbf{v}_i^n + \frac{1}{2} (\mathbf{a}_i^{n+1} + \mathbf{a}_i^n) \Delta t \\ \mathbf{u}_i^{n+1} = \mathbf{u}_i^n + \mathbf{v}_i^n \Delta t + \frac{1}{2} \mathbf{a}_i^n \Delta t^2 \\ \mathbf{x}_i^{n+1} = \mathbf{x}_i^n + \mathbf{u}_i^{n+1} \end{cases}$$

Courant-Friedrichs-Levy stability condition

$$\Delta t < \min \left(\frac{h_{\text{smooth}}}{|\mathbf{v}_{\text{max}}| + c_B}, \frac{1}{8} \frac{h_{\text{smooth}}^2}{\max(\mu/\rho)} \cdot \frac{1}{4} \sqrt{\frac{h_{\text{smooth}}}{g}}, \frac{1}{4} \sqrt{\frac{\min(\rho_0) h^3}{2\pi\beta}} \right)$$

The non-local form of the surface tension formula



$$s(\mathbf{x}') - s(\mathbf{x}) = \begin{cases} \frac{2\rho_{x,0}}{\rho_{x,0} + \rho_{x',0}}, & \text{if } \mathbf{x} \text{ and } \mathbf{x}' \text{ belong to different phases} \\ 0, & \text{if } \mathbf{x} \text{ and } \mathbf{x}' \text{ belong to the same phase} \end{cases}$$

Introduce the truncation function to avoid the unit normal vector field having incorrect directions

$$\mathbf{n}(\mathbf{x}) = \begin{cases} \frac{\int_{H_K} (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\xi) dV_{x'}}{\int_{H_K} (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\xi) dV_{x'}}, & \text{if } \|\nabla s\| > 1.0 \times 10^{-2} \Delta x \\ 0, & \text{else} \end{cases}$$

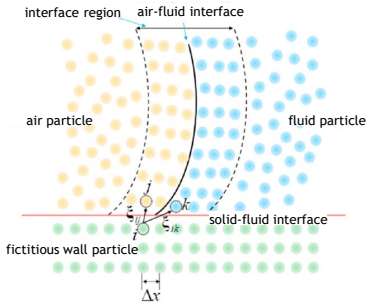
$$\kappa^*(\mathbf{x}) = -\nabla \cdot \mathbf{n}(\mathbf{x}) = -\int_{H_K} (\mathbf{n}(\mathbf{x}') - \mathbf{n}(\mathbf{x})) \cdot \mathbf{g}_1(\xi) dV_{x'}$$

Introduce correction factors to correct the curvature field of boundary particles let $\mathbb{N}(\mathbf{x}) = \begin{cases} 1, & \text{if } \|\nabla s\| > 1.0 \times 10^{-2} \Delta x \\ 0, & \text{else} \end{cases}$

$$\zeta^*(\mathbf{x}) = \frac{\int_{H_K} \min(\mathbb{N}(\mathbf{x}), \mathbb{N}(\mathbf{x}')) w_0(\|\xi\|) dV_{x'}}{\int_{H_K} w_0(\|\xi\|) dV_{x'}}, \kappa^{**}(\mathbf{x}) = \frac{\kappa^*(\mathbf{x})}{\zeta^*(\mathbf{x})}$$

(Gao et al., 2020)

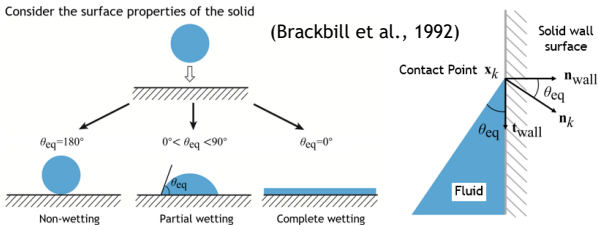
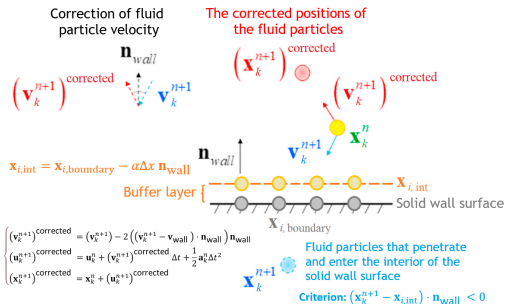
边界条件



Considering the no-slip boundary, the fluid velocity is zero at the wall surface.

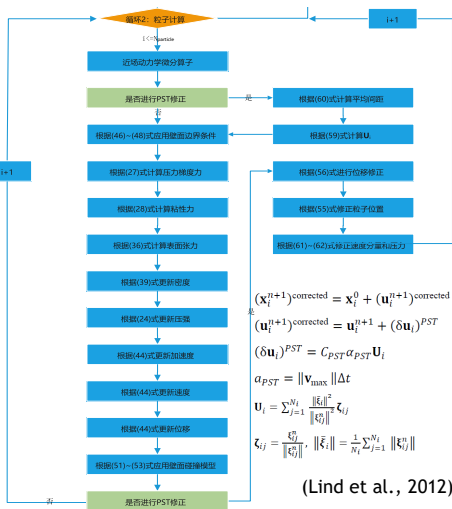
$$\begin{aligned} \mathbf{v}_i^n &= 2\mathbf{v}_{\text{wall}} - \sum_{k=1}^{N_f} w_0(\|\xi_{ij}^n\|) \mathbf{w}_k^n \\ P_i &= \frac{\sum_{k=1}^{N_f} (P_k + (\mathbf{F}_k^B - p_k \mathbf{a}_{\text{wall}}) \cdot \xi_{ij}^n) w_0(\|\xi_{ij}^n\|)}{\sum_{k=1}^{N_f} w_0(\|\xi_{ij}^n\|)} \\ \rho_i &= \rho_{0,i} \left(\frac{P_i - P_{0,i}}{P_{\text{ref},i}} + 1 \right)^{1/\gamma_i} \end{aligned}$$

(Gao et al., 2020)



防聚集的粒子迁移技术（Particle Shifting Technology）

- PST技术通过引入位移修正来调整粒子位置，使粒子在移动后保持近似均匀分布



$$\begin{aligned}
 (\mathbf{v}_i^{n+1})^{\text{corrected}} = & \left[\mathbf{v}_{i1}^{n+1} + \frac{2\Delta x^2}{\pi\delta^2} \sum_{j=1}^{N_i} (\mathbf{v}_{j1}^n - \mathbf{v}_{i1}^n) \frac{[\xi_{ij1} \quad \xi_{ij2}]^T}{\|\xi_{ij1}^2 + \xi_{ij2}^2\|} \cdot (\delta \mathbf{u}_i)^{PST} \right. \\
 & + \frac{3\Delta x^2}{2\pi\delta^3} \sum_{j=1}^{N_i} ((\delta \mathbf{u}_i)^{PST})^T \cdot (\mathbf{v}_{j1}^n - \mathbf{v}_{i1}^n) \frac{\begin{pmatrix} 3\xi_{i1}^2 - \xi_{i2}^2 & 4\xi_{i1}\xi_{i2} \\ 4\xi_{i1}\xi_{i2} & -\xi_{i1}^2 + 3\xi_{i2}^2 \end{pmatrix}}{\|\xi_{ij1}^2 + \xi_{ij2}^2\|^{3/2}} \cdot (\delta \mathbf{u}_i)^{PST}, \\
 & \mathbf{v}_{i2}^{n+1} + \frac{2\Delta x^2}{\pi\delta^2} \sum_{j=1}^{N_i} (\mathbf{v}_{j2}^n - \mathbf{v}_{i2}^n) \frac{[\xi_{ij1} \quad \xi_{ij2}]^T}{\|\xi_{ij1}^2 + \xi_{ij2}^2\|} \cdot (\delta \mathbf{u}_i)^{PST} \\
 & \left. + \frac{3\Delta x^2}{2\pi\delta^3} \sum_{j=1}^{N_i} ((\delta \mathbf{u}_i)^{PST})^T \cdot (\mathbf{v}_{j2}^n - \mathbf{v}_{i2}^n) \frac{\begin{pmatrix} 3\xi_{i1}^2 - \xi_{i2}^2 & 4\xi_{i1}\xi_{i2} \\ 4\xi_{i1}\xi_{i2} & -\xi_{i1}^2 + 3\xi_{i2}^2 \end{pmatrix}}{\|\xi_{ij1}^2 + \xi_{ij2}^2\|^{3/2}} \cdot (\delta \mathbf{u}_i)^{PST} \right]^T
 \end{aligned}$$

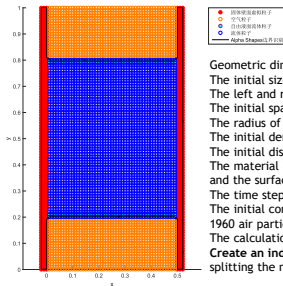
$$\begin{aligned}
 (\mathbf{x}_i^{n+1})^{\text{corrected}} = & \mathbf{x}_i^0 + (\mathbf{u}_i^{n+1})^{\text{corrected}} \\
 (\mathbf{u}_i^{n+1})^{\text{corrected}} = & \mathbf{u}_i^{n+1} + (\delta \mathbf{u}_i)^{PST} \\
 (\delta \mathbf{u}_i)^{PST} = & C_{PST} \alpha_{PST} \mathbf{U}_i \\
 \alpha_{PST} = & \|\mathbf{v}_{\max}\| \Delta t \\
 \mathbf{U}_i = & \sum_{j=1}^{N_i} \frac{\|\tilde{\mathbf{k}}_i\|^2}{\|\xi_{ij}\|^2} \zeta_{ij} \\
 \zeta_{ij} = & \frac{\xi_{ij}^n}{\|\xi_{ij}\|}, \quad \|\tilde{\mathbf{k}}_i\| = \frac{1}{N_i} \sum_{j=1}^{N_i} \|\xi_{ij}^n\|
 \end{aligned}$$

速度和压力也可以根据位移的修正量进行修正
(二维完整对称近场域下相对值形式)

(Lind et al., 2012)

(Gao et al., 2020)

数值实现（算例设定、近场域搜索、自由液面识别）



Geometric dimensions: $0.5 \text{ cm} \times 1.0 \text{ cm}$.

The initial size of the central liquid bridge is $0.5 \text{ cm} \times 0.6 \text{ cm}$.

The left and right boundaries are stationary rigid solid walls.

The initial spacing between the material points $\Delta x = 0.01 \text{ cm}$.

The radius of the near-field domain $\delta = 3.015 \Delta x$.

The initial density of the fluid particles is set to $\rho_0 = 1 \text{ kg/m}^3$.

The initial displacement, velocity, acceleration and pressure are all zero.

The material constant of the fluid $\gamma = 1$, the artificial sound speed $c_f = 12 \text{ cm/s}$, the dynamic viscosity coefficient $\mu = 0.2 \text{ Ps}\cdot\text{s}$, and the surface tension coefficient $\beta = 1 \text{ N/m}$.

The time step $\Delta t = 5.0 \times 10^{-7} \text{ s}$, and the total number of time steps $N = 15000$.

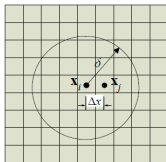
The initial computational domain was discretized into 2989 fluid particles, 606 fictitious particles representing solid walls, and 1960 air particles.

The calculation program for the example was written using **MATLAB 2015b**.

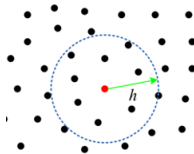
Create an independent main program file that encompasses all the calculation processes, rather than adopting the method of splitting the main program into multiple sub-program scripts and then calling them.



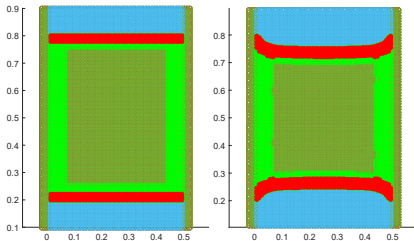
<https://github.com/Liskelleo/Numerical-Simulation-of-the-Evolution-of-Water-Meniscus-Based-on-Peridynamics>



Two-dimensional structure with central orthogonal uniform discretization.

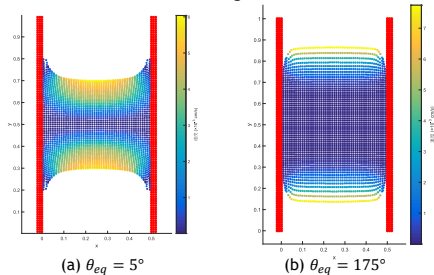


By calling the library function rangesearch to conduct a KD-tree neighborhood search for the specified particle.

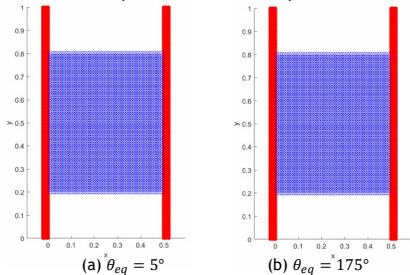


计算结果

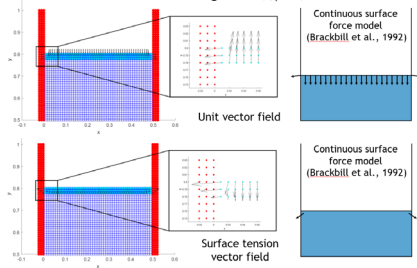
Final configuration



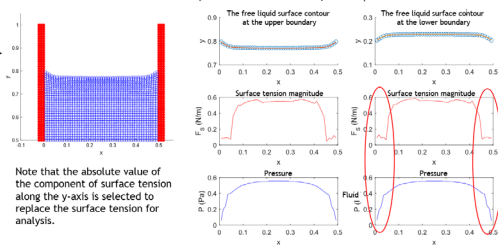
Liquid level evolution process



Initial configuration ($\theta_{eq} = 5^\circ$)

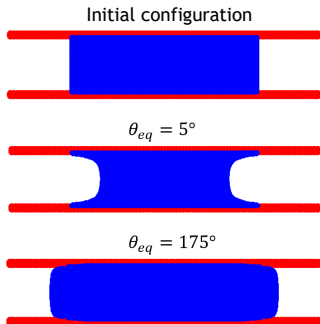


Particle distribution at 7500 time steps for the concave liquid surface example ($\theta_{eq} = 5^\circ$)

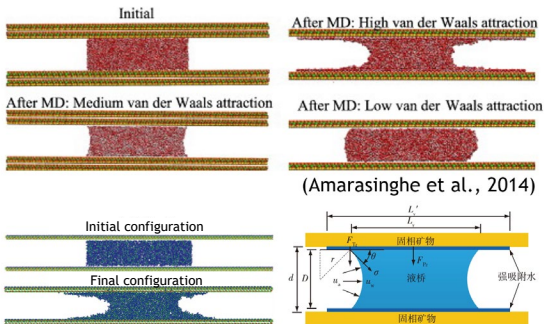


算例比较 (PD vs. MD)

The stronger the adsorption of the solid boundary on the pore water, the more "concave" the curved liquid surface will be; conversely, it will be more "convex".



(a) PD results



(Luo et al., 2022)

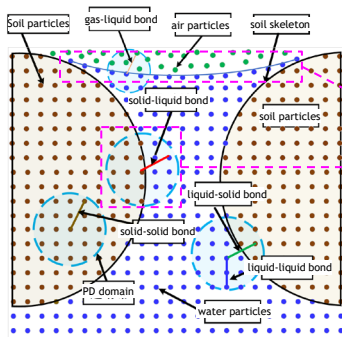
(b) MD results

“热-水-力”耦合作用下弯液面的演化探究（未来工作）

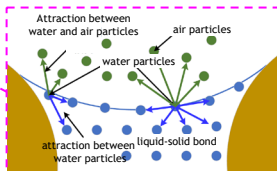
Simplified single liquid bridge model

Capillary action ✓
Evaporation effect ✗

Investigation on the Evolution of Curved Liquid Surface under the Coupling Effect of Heat, Water and Force is our future work!



The microscopic structure of clay

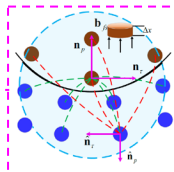


One-way gas-liquid interaction

Boundary conditions:

Gas $\rightarrow$ Liquid: Surface tension

Solid $\rightleftharpoons$ Liquid: Decomposes into two forces acting in the pressure direction and the shear direction



Fluid-solid interaction